

# Where explanation guidance helps, and where it does not

Matched test images and generated answer excerpts; blue text marks the guided student.

## Improved visual grounding



**Local texture and shape**  
**Saliency preserves small discriminative surface details.**

**Student, global MSE**  
"The image shows a close-up of a mushroom with a distinctive appearance. The cap of the mushroom is light brown and has a slightly textured surface, with some areas appearing more raised or bumpy than others ..."

**Reading** The saliency-guided answer keeps the cap's raised texture and reddish surface, while the global-MSE student drifts toward a generic mushroom description.

**Student, explanation-guided**  
"The image shows a close-up of a mushroom with a reddish-brown cap. The cap has a rough, textured surface with small, raised spots and ridges. The edges of the cap appear to be slightly curled inward. The ..."



**Object identity recovery**  
**The guided student recovers the visible object identity.**

**Student, global MSE**  
"The image shows a pair of red high-heeled shoes with white trim and a pointed toe. The shoes are placed on a wooden surface, and there is a patterned wallpaper in the background featuring a mix of geometric shapes ..."

**Reading** The explanation-guided output describes bowls and colors, whereas the MSE student produces an unrelated shoe description.

**Student, explanation-guided**  
"The image shows three bowls placed on a surface. The bowls are of different colors: one is red, another is brown, and the third is a combination of both colors. They appear to be ceramic or porcelain, with smooth ..."

## Remaining limitations



**Fine-grained semantic failure**  
**Local visual evidence is not enough for full scene semantics.**

**Teacher VLM**  
"The image shows a person standing next to a large, detailed dinosaur skull mounted on a stand. The skull appears to be that of a Triceratops, characterized by its three prominent horns and frill. The person is ..."

**Reading** The student captures the Triceratops skull but misses broader context, including the nearby person and richer scene description.

**Student, explanation-guided**  
"The image shows a close-up of a dinosaur skull, specifically that of a Triceratops. The skull is displayed in a museum setting, with a window in the background allowing natural light to illuminate the ..."



**Generation robustness failure**  
**The plain MSE visual swap can break generation.**

**Student, global MSE**  
"I'm sorry, but I cannot see any image attached to your message. Could you please provide more information or rephrase your question?"

**Reading** Explanation guidance improves visual-path usability: the guided answer describes the skeleton, while the MSE student says it cannot see the image.

**Student, explanation-guided**  
"The image depicts a dinosaur skeleton on display in what appears to be a museum or exhibition hall. The skeleton is mounted on a stand and is positioned in a standing posture, with its legs spread apart and arms ..."